

NEWTON, KS, USA

WMO: 724509

Lat: 38.068N

Lon: 97.275W

Elev: 467

StdP: 95.84

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53939

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.1	-12.5	-20.8	0.6	-13.1	-17.8	0.8	-10.9	14.6	3.1	13.0	2.9	4.9	0	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	37.4	23.0	35.2	23.0	33.0	22.6	25.1	32.2	24.4	31.4	23.8	30.7	6.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.8	28.1	22.5	18.2	27.7	22.1	17.7	27.3	79.5	32.1	76.3	31.8	73.7	30.8	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.0	11.8	10.7	DB	-17.8	39.7	3.0	1.5	-20.0	40.8	-21.7	41.7	-23.4	42.5	-25.5	43.6	
			WB	-18.4	26.4	3.2	0.9	-20.7	27.1	-22.6	27.6	-24.3	28.1	-26.6	28.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	-0.5	1.9	7.3	12.5	18.1	23.8	26.4	25.4	20.8	14.3	7.0	0.9
	DBStd	10.69	6.15	6.45	6.43	5.33	4.63	3.57	3.41	3.34	4.36	5.08	5.53	5.71
	HDD10.0	1146	328	233	127	34	2	0	0	0	1	18	120	283
	HDD18.3	2688	585	459	345	184	63	4	0	1	24	144	340	539
	CDD10.0	2321	1	8	42	109	253	414	507	478	325	151	31	2
	CDD18.3	821	0	0	2	10	56	168	249	219	98	18	1	0
	CDH23.3	8544	0	1	14	107	496	1689	2832	2309	922	170	5	0
	CDH26.7	3926	0	0	2	21	159	749	1459	1133	362	41	1	0
Wind	WSAvg	5.1	5.2	5.3	5.8	6.0	5.1	5.0	4.5	4.2	4.5	5.0	5.3	5.0
Precipitation	PrecAvg	862	22	31	64	77	118	133	106	90	78	68	47	26
	PrecMax	1176	77	97	139	174	288	260	323	194	152	183	209	91
	PrecMin	637	2	0	8	31	21	66	25	6	11	6	2	2
	PrecStd	136	18	25	33	40	66	43	67	45	43	49	51	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	21.2	25.4	28.9	33.0	37.3	39.8	38.9	36.0	30.9	23.8	17.6
		MCWB	9.4	12.0	14.9	18.2	20.8	23.0	22.8	22.9	22.3	20.5	14.7	12.1
	2%	DB	13.8	17.6	22.4	26.3	30.9	34.3	37.6	37.2	32.6	27.4	21.2	14.2
		MCWB	7.4	10.5	14.5	16.5	20.9	22.9	23.1	23.0	21.9	17.9	13.6	9.0
	5%	DB	11.3	14.7	19.9	23.8	28.0	32.8	36.0	34.7	31.0	25.0	17.9	12.2
		MCWB	6.4	9.3	12.9	16.0	19.9	22.4	23.3	22.9	21.6	17.1	12.0	8.0
	10%	DB	8.5	12.1	17.5	22.0	26.3	31.2	33.6	32.5	28.6	22.5	16.0	9.1
		MCWB	4.8	7.9	12.0	15.1	19.2	22.0	23.1	22.6	20.3	16.4	10.9	5.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	14.2	18.0	20.2	23.5	25.6	25.9	26.2	24.4	22.4	16.7	14.4
		MCDB	14.3	18.2	22.9	25.7	29.6	32.5	33.1	32.5	30.7	26.8	20.7	16.3
	2%	WB	8.6	12.0	15.9	18.8	22.2	24.6	25.1	25.1	23.3	20.6	14.8	10.5
		MCDB	13.0	15.3	19.7	24.0	28.2	31.4	32.5	31.7	29.7	24.6	18.7	12.6
	5%	WB	6.8	10.0	14.4	17.4	21.3	23.7	24.4	24.2	22.4	18.9	13.4	7.9
		MCDB	10.9	14.8	18.6	22.0	26.6	30.3	31.8	30.8	28.6	23.2	17.4	11.3
	10%	WB	4.7	7.3	12.1	15.8	20.1	22.9	23.8	23.5	21.5	17.1	11.6	5.8
		MCDB	8.7	11.7	17.1	20.8	24.9	29.3	31.1	30.3	27.1	21.4	15.4	9.5
Mean Daily Temperature Range	5% DB	MDBR	10.8	11.2	12.0	12.0	11.2	11.5	11.7	11.7	11.8	11.9	11.2	9.9
		MCDBR	15.1	15.5	15.4	15.0	13.4	13.4	14.3	14.5	13.8	14.5	14.4	13.6
		MCWBR	9.9	10.1	9.3	8.8	6.6	5.0	4.5	4.8	5.5	7.5	8.6	9.3
	5% WB	MCDBR	14.3	14.4	12.4	13.0	11.7	11.9	12.1	11.7	12.1	11.8	12.2	12.3
		MCWBR	9.9	10.1	8.7	8.7	6.9	5.5	4.8	5.2	5.6	7.2	8.4	9.3
Clear-Sky Solar Irradiance	taub	0.282	0.293	0.326	0.370	0.404	0.413	0.420	0.422	0.378	0.334	0.296	0.279	
	taud	2.512	2.485	2.425	2.322	2.300	2.305	2.326	2.289	2.378	2.488	2.522	2.521	
	Ebn at Noon	913	943	937	908	879	867	858	847	869	883	886	891	
	Edh at Noon	77	90	106	125	130	130	126	128	110	89	76	71	
All-Sky Solar Radiation	RadAvg	2.47	3.20	4.21	5.30	5.99	6.81	6.70	6.00	4.96	3.66	2.70	2.12	
	RadStd	0.24	0.43	0.29	0.34	0.50	0.44	0.42	0.33	0.33	0.41	0.25	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page